

Research Proposal

Partition Geometry as a Predictive Framework for Soft Matter Rheology and Differential Dynamic Microscopy

Kundai Farai Sachikonye — Application for predoctoral position, SoMeX Lab (Job-ID 5570)

Central Question

Can the rheological properties of yield stress fluids and the intermediate scattering functions measured by Differential Dynamic Microscopy (DDM) be predicted from first principles — without empirical fitting — using a single unified framework based on bounded phase space geometry?

Background and Motivation

Yield stress fluids (concentrated emulsions, colloidal gels, polymer networks, biological hydrogels) are among the most practically important and theoretically challenging soft matter systems. Their defining property — the existence of a critical stress σ_y below which the material behaves as a solid and above which it flows — is typically treated phenomenologically: the Herschel-Bulkley or Bingham models fit experimental flow curves but provide no prediction of σ_y from molecular or colloidal parameters.

DDM offers a powerful experimental route to the dynamic properties of such systems: the image structure function $D(\mathbf{q}, \Delta t)$ gives access to the intermediate scattering function $f(\mathbf{q}, \Delta t)$ without the sample constraints of DLS, enabling in-situ characterisation of structure and dynamics during flow. Yet the interpretation of DDM data in yield stress fluids remains challenging: the spatial and temporal heterogeneity of these materials, and the optical complexity of concentrated colloidal dispersions, make connecting $D(\mathbf{q}, \Delta t)$ to mechanical properties non-trivial.

Proposed Framework

I propose to develop and test a first-principles connection between the rheological behaviour and DDM-measured dynamics of model yield stress systems, using the bounded phase space (BPS) framework I have developed.

Step 1 — Rheological predictions from partition geometry. In the BPS framework, every soft matter system is a bounded oscillatory system characterised by S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$. Viscosity is $\mu = \tau_c \times g$, where τ_c is the partition lag (timescale of spectral mode transitions) and g is the coupling strength (off-diagonal elements of the spectral coherence matrix). The yield stress σ_y corresponds to the partition floor: the minimum S-entropy coordinate $S_e^{\min} = 1/m > 0$ (Floor Theorem, where m is the number of spectral modes), below which the ordered phase cannot sustain coherence. This gives a structural prediction: σ_y is proportional to the spectral coherence floor of the colloidal assembly, which can be computed from particle size distribution and volume fraction before any mechanical measurement.

Step 2 — DDM structure function from optical partition geometry. The image

structure function $D(\mathbf{q}, \Delta t) = 2[S(\mathbf{q}) - \text{Re}\{S(\mathbf{q}, \Delta t)\}]$, where $S(\mathbf{q}, \Delta t)$ is the dynamic structure factor. In the BPS framework, the optical signal in a DDM experiment is governed by the Resonant Stack theorem: the transmittance $T = 1/e$ is the universal fixed point of the Beer-Lambert-Kirchhoff equivalence, and the image contrast encoding particle dynamics is a projection of the partition structure function onto the imaging wave vector $\mathbf{q}$. This gives a predictive model for $D(\mathbf{q}, \Delta t)$ from the same partition geometry that predicts μ and σ_y — a single framework connecting rheology to DDM without separate fitting.

Step 3 — Experimental validation in the SoMeX Lab. The proposal would be tested against measurements on two model systems: (i) Carbopol microgel suspensions (archetypal soft yield stress fluid; well-characterised by the lab’s Anton Paar MCR 702e and ShearView rheometer); (ii) hard-sphere colloidal suspensions near the glass transition (DDM-accessible via the lab’s sCMOS camera systems and fastDDM code). For each system, the BPS framework predicts σ_y and the shape of $D(\mathbf{q}, \Delta t)$ from the particle parameters alone. Agreement or disagreement with measurement is a direct test of whether the partition geometry captures the relevant physics.

Expected Outcomes and Timeline

Year 1: Implement BPS rheology predictions for Carbopol and hard-sphere systems; compare to ShearView measurements; extend fastDDM analysis to extract structure functions for BPS validation.

Year 2–3: Extend to biologically relevant yield stress systems (mucus, cytoskeletal networks, bacterial biofilm matrices) and to active matter DDM (bacterial suspensions), where the $\oint \mathbf{J} \cdot d\mathbf{A} = 0$ conservation law for collective motion provides an additional first-principles constraint on $f(\mathbf{q}, \Delta t)$.

The outcome, if the predictions hold, would be a validated first-principles connection between particle-level structure and macroscopic rheological and optical observables in soft matter systems — a contribution both to fundamental soft matter physics and to the practical characterisation of biological fluids and active matter.